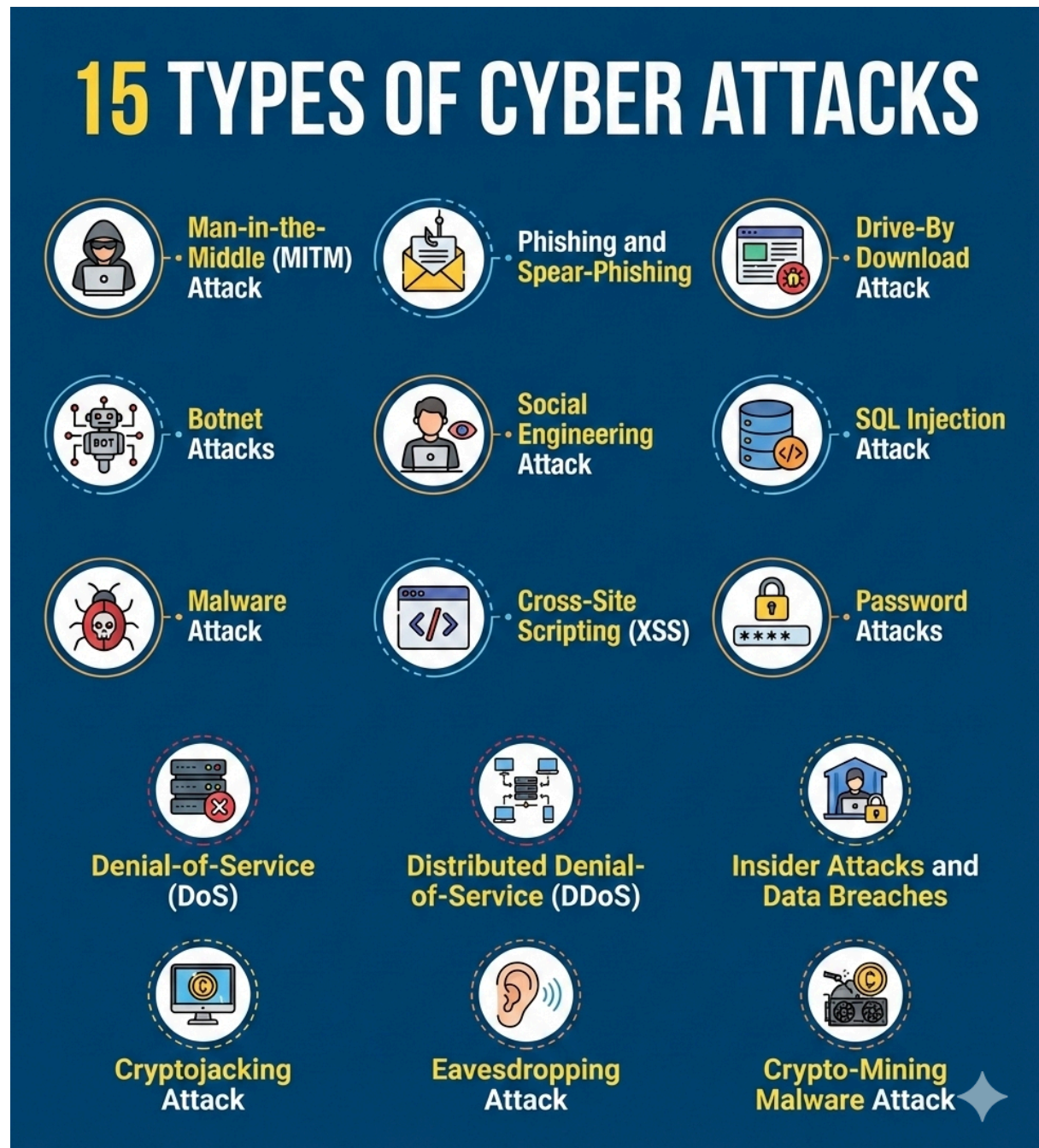


# 15 Types of Cyber Attacks (With Real-World Examples)



Cyber attacks continue to evolve rapidly. Understanding **how they work, where they occur in real scenarios, and how to defend against them** is essential for cybersecurity professionals.

This report covers 15 major cyber attack types including definitions, mechanisms, examples, impacts, and mitigations.

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## 1. Man-in-the-Middle (MITM) Attack

### Definition:

A MITM attack occurs when a threat actor intercepts communication between two parties to steal or manipulate data.

### Example:

A user connects to an unsecured public Wi-Fi in a café. An attacker intercepts login credentials to online banking during the session.

### Impact:

- Credential theft
- Data modification
- Session hijacking

### Prevention:

- HTTPS enforcement
  - VPN usage
  - TLS certificate pinning
- 

## 2. Phishing and Spear-Phishing

### Definition:

Deceptive messages that trick users into clicking malicious links, downloading malware, or revealing sensitive information.

### Example:

A fake Microsoft Office 365 login page emailed to employees asking them to "reset their password."

### Impact:

- Account takeover
- Malware infection
- Financial fraud

**Prevention:**

- Email filtering
  - User awareness training
  - DMARC, SPF, and DKIM policies
- 

### 3. Drive-By Download Attack

**Definition:**

Malicious code is automatically downloaded when visiting a compromised or malicious website.

**Example:**

A user visits a hacked blog containing exploit kits that silently install ransomware.

**Impact:**

- Malware infection
- System compromise

**Prevention:**

- Web filtering
  - Frequent software patching
  - Browser hardening
- 

### 4. Botnet Attacks

**Definition:**

A network of compromised devices remotely controlled by an attacker.

**Example:**

The Mirai botnet hijacked millions of IoT devices and launched large-scale DDoS attacks.

**Impact:**

- Massive DDoS attacks
- Credential abuse

**Prevention:**

- Network anomaly detection
  - Patch insecure IoT devices
  - Endpoint protection agents
- 

## 5. Social Engineering Attack

**Definition:**

Psychological manipulation to trick users into sharing confidential information or granting access.

**Example:**

An attacker impersonates IT support and convinces an employee to reveal their MFA code.

**Impact:**

- Unauthorized access
- Data breaches

**Prevention:**

- Zero trust behavior
  - Security training
  - Identity verification procedures
- 

## 6. SQL Injection Attack

**Definition:**

Malicious SQL queries inserted into input fields to access or manipulate a backend database.

**Example:**

Entering `' OR '1'='1` into a login form bypasses authentication on an insecure web application.

**Impact:**

- Credential theft
- Full database extraction

**Prevention:**

- Parameterized queries
  - Input validation
  - Web Application Firewalls
- 

## 7. Malware Attack

**Definition:**

Malicious software designed to harm, exploit, or steal from systems.

**Example:**

The WannaCry ransomware outbreak infected over 200,000 systems by exploiting SMB vulnerabilities.

**Impact:**

- Data loss
- System downtime
- Financial extortion

**Prevention:**

- Endpoint security
  - Network segmentation
  - Patch management
- 

## 8. Cross-Site Scripting (XSS)

**Definition:**

Injection of malicious JavaScript into trusted websites to execute scripts in a victim's browser.

**Example:**

A comment box on a blog executes malicious JavaScript such as

```
<script>document.cookie</script>
```

 exposing session tokens.**Impact:**

- Account takeover
- Redirect attacks

**Prevention:**

- Output encoding
  - CSP implementation
  - Sanitizing input
- 

## 9. Password Attacks

**Definition:**

Attempts to crack or obtain passwords to gain unauthorized access.

**Example:**

Attackers perform credential stuffing using leaked credentials from a past breach to log into other accounts.

**Impact:**

- Account compromise
- Unauthorized network access

**Prevention:**

- MFA enforcement
  - Strong password policies
  - Rate limiting
- 

## 10. Denial-of-Service (DoS) Attack

**Definition:**

A single attacker overwhelms a system or network with excessive requests.

**Example:**

A single host sends floods of TCP SYN packets to a web server causing service disruption.

**Impact:**

- Unavailability of services
- Financial loss

**Prevention:**

- Rate limiting
  - Firewall filtering
  - Resource throttling
- 

## 11. Distributed Denial-of-Service (DDoS) Attack

**Definition:**

A coordinated attack from multiple infected devices or botnets.

**Example:**

The 2016 Dyn DNS attack disrupted Twitter, Netflix, GitHub, and Spotify.

**Impact:**

- Large-scale outages
- Brand damage

**Prevention:**

- Cloud DDoS protection
  - Load balancing
  - Traffic scrubbing solutions
- 

## 12. Insider Attacks and Data Breaches

**Definition:**

Security incidents caused intentionally or accidentally by employees or trusted parties.

**Example:**

A disgruntled employee exports customer PII to a USB drive and sells it to competitors.

**Impact:**

- Intellectual property theft
- Legal consequences

**Prevention:**

- Access control rules
  - Activity logging
  - Zero trust model
- 

## 13. Cryptojacking Attack

**Definition:**

Unauthorized use of a victim's computing resources to mine cryptocurrency.

**Example:**

A malicious JavaScript mining script embedded on websites uses visitors' CPU to mine Monero.

**Impact:**

- Performance degradation
- Increased operating costs

**Prevention:**

- Browser script blocking
  - Endpoint behavior monitoring
- 

## 14. Eavesdropping Attack

**Definition:**

Passive interception of private communication traffic.

**Example:**

Using a packet sniffer like Wireshark on an unsecured wireless network to capture credentials.



**Impact:**

- Confidential data leaks
- Privacy violations

**Prevention:**

- Encrypted communications
  - Secure network configurations
- 

## 15. Crypto-Mining Malware Attack

**Definition:**

Executable malware specifically designed to mine cryptocurrency.

**Example:**

Malware installed via malicious software cracks or pirated downloads hijacks GPU power for mining.

**Impact:**

- Hardware damage
- System slowdown

**Prevention:**

- EDR solutions
  - Application whitelisting
  - Threat hunting
- 

## Summary

These attack types represent common and emerging security threats.

Understanding **how they occur, real-world examples, impacts, and defense mechanisms** helps organizations and professionals strengthen prevention and detection strategies.

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